

Using AI/ML and Remote Sensing to detect potential sources of waterbody pollution

River Deep Mountain AI

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Abstract—Agricultural sources of pollution have a considerable and persistent impact on water quality. Water quality monitoring and catchment walkovers for detecting agricultural pollution are time consuming and expensive. Consequently, these traditional methods have seen limited application across space and time and often struggle to capture rapid land-use changes and to pinpoint high pollution risk areas. The River Deep Mountain AI (RDMAI) initiative aims to address this gap by introducing three open-source AI/ML and Remote Sensing models for detecting three sources of agricultural pollution - Bare Cropland Detection, Slurry Tank Detection, and Poached Land Detection. Together, these models create a scalable, cloud-based geospatial intelligence framework that empowers regulators, water companies, and researchers to monitor and manage agricultural pollution risks proactively.

Index Terms—Water quality, Slurry Tank Detection, Bare Cropland Detection, Poached Land Detection, Machine Learning, Environmental Monitoring, AI

I. INTRODUCTION

In agricultural landscapes, both point and diffuse pollution sources impact catchment water quality. For instance, poorly maintained slurry storage (e.g. tanks) can leak nutrients and organic matter into nearby waterways, representing a clear point-source risk, and the presence of slurry tanks is one potential indicator of intensive livestock production and the pollution risks associated with livestock grazing and manure application. Meanwhile, bare croplands and poached lands can exacerbate diffuse pollution by increasing the risk of sediment and nutrient runoff during rainfall events. Physical properties, such as soil type and slope, contribute to complex patterns of agricultural pollution. In addition, traditional field surveys, such as water quality monitoring and catchment walkovers, are expensive and time consuming, leading to their limited application. Consequently, these traditional methods are often unable to capture the full spatial and temporal dynamics in agricultural pollution.

Existing land cover datasets are annual products that do not explicitly capture over-winter cover. Therefore, these datasets do not fully capture the spatial and temporal extent of bare soil. In addition, they are not available in near-real time. Also, there are currently no datasets showing poaching or the location of manure management facilities.

RDMAI offers open, transparent, and scientifically validated models identifying bare soil, poaching and slurry tanks that integrate seamlessly with GIS and catchment dashboards for consistent, nationwide application.

II. MODEL HIGHLIGHTS

A. Open Slurry Tank Detection Model

Understanding the location of agricultural slurry tanks is important for assessing agricultural sources of pollution. First, they act as an indicator of intensive livestock production and the pollution risks associated with livestock grazing and manure application. Second, poorly maintained slurry tanks can leak nutrients and organic matter into nearby surface water and groundwater. Therefore, slurry tanks are indicators of both potential point and diffuse sources of agricultural pollution.

The Open Slurry Tank Detection Model identifies agricultural circular above-ground slurry storage tanks using high-resolution satellite imagery. Using a YOLOv8 deep learning pipeline (Fig. 1) trained on public and custom datasets, the model can generate geoslurry tank detections that can be imported into GIS.

Use Case: Current catchment management approaches rely on time-consuming, in-person walkovers or manual searching through aerial imagery to locate and assess the risk that slurry tanks and the associated livestock potentially pose to waterbody health. The Open Slurry Tank Detection Model provides stakeholders with a method for automatically, efficiently, and rapidly locating slurry tanks over large catchment areas. The Open Slurry Tank Detection Model can be used in conjunction with existing tools (e.g. rainfall-runoff models, geospatial datasets, etc.) to identify areas at greater risk of potential agricultural water quality pollution, thus allowing for walkovers to be targeted at these potentially higher risk areas.

B. Open Bare Cropland Detection Model

The Open Bare Cropland Detection Model identifies crop fields left bare or ‘fallow’. Bare fields are at higher risk of erosion and nutrient runoff and are a potential source of agricultural diffuse pollution. Using Sentinel-2 imagery and spectral indices such as Normalized Difference Vegetation Index (NDVI), Soil Adjusted Vegetation Index (SAVI), Bare Soil

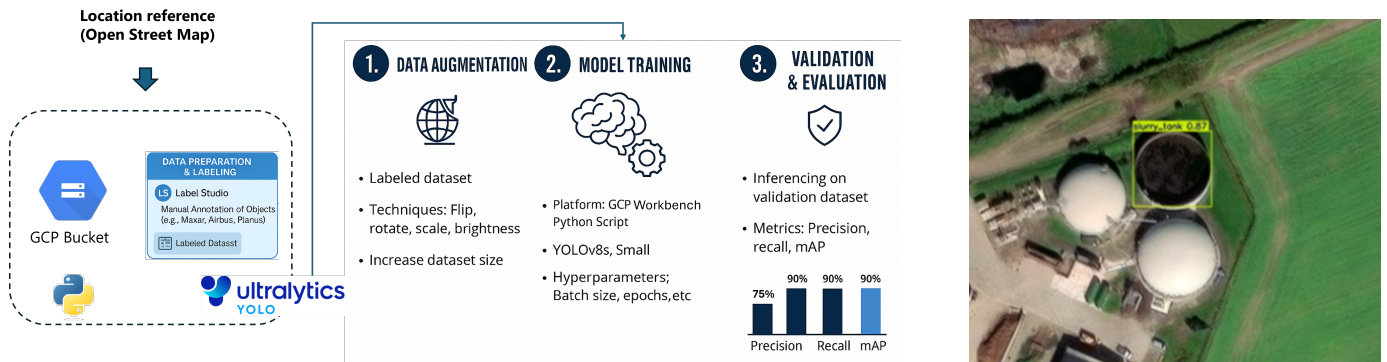


Fig. 1: Methodology and output of the RDMAI Open Slurry Tank Detection Model

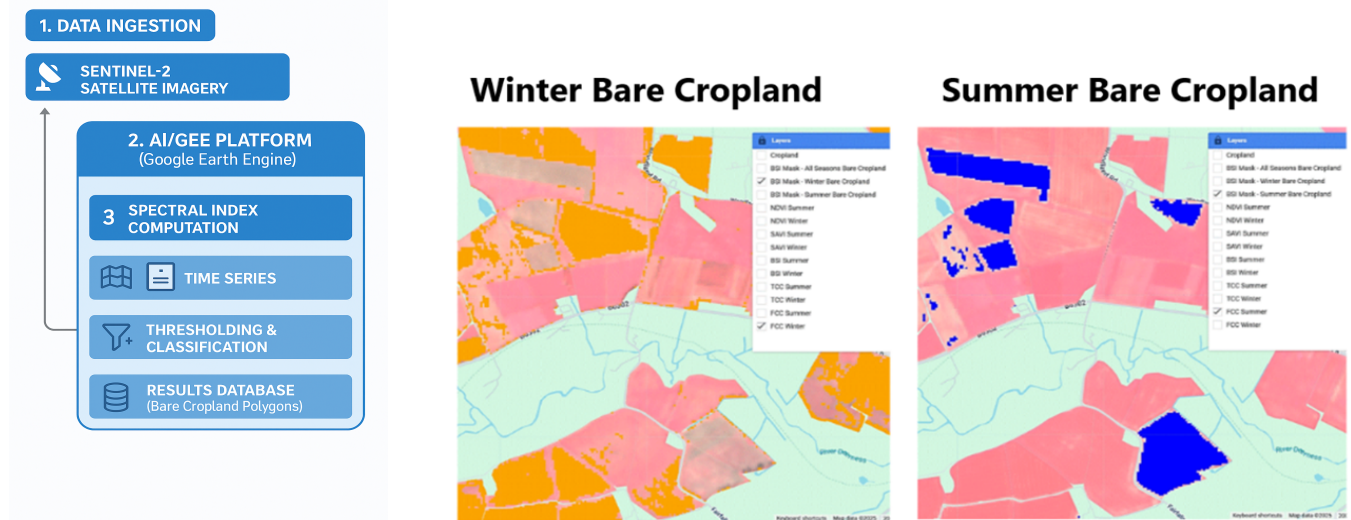


Fig. 2: Methodology and output of the RDMAI Open Bare Cropland Detection Model



Fig. 3: Methodology and output of the RDMAI Open Poached Land Detection Model

index (BSI), Normalized Difference Moisture Index (NDMI), and Normalized Difference Snow index (NDSI) within Google Earth Engine, the model identifies seasonal soil exposure and generates time series outputs to support evidence-based catchment management (Fig. 2). The tool allows stakeholders to understand how areas of bare cropland change within a catchment at sub-annual timescales, enabling more effective targeting of catchment interventions that can reduce the risk of soil erosion and nutrient runoff.

Use Case: This model can identify areas of bare land during winter and summer. These identified areas of land can then be used in conjunction with existing tools (e.g. rainfall runoff models) to identify key areas of risk within a catchment.

C. Open Poached Land Detection Model

The Open Poached Land Detection Model identifies areas of degraded soils caused by livestock trampling or farm vehicle tracks. Poached land is at greater risk of runoff, erosion, and sediment transport. Therefore, it is a potential source of agricultural diffuse pollution.

The outputs of the model are georeferenced identifications of poached land that can be imported into GIS (Fig. 3).

Use Case: The outputs of the Open Poached Land Detection Model could be used in conjunction with existing tools (e.g. rainfall-runoff models) to highlight key areas of runoff pollution risk within a catchment. This can enable stakeholders to make more informed decisions on what and where particular interventions would be most effective within a catchment.

III. DATA SOURCES

The datasets used to build the models, whether in Google Earth Engine or the Google Cloud Platform, are shown in Table 1.

IV. MODEL VALIDATION AND ACCURACY

A. Open Bare Cropland Detection Model

To ensure consistency across diverse landscapes, validation of the Open Bare Cropland Detection Model was conducted using an out-of-sample dataset covering 19 UK catchments that spanned four dominant soil types (Chalky, Clayey, Peat, and Sandy, as defined by the British Geological Survey in their soil type dataset). In the absence of ground-truth data, Google Earth Pro's high-resolution historical imagery was used for manual visual verification across 2,094 fields, ensuring an unbiased assessment of model performance under varied soil and climatic conditions. This robust validation confirmed that the model performs effectively across diverse agricultural and geopedological settings.

As shown in Table 2, analysis across four major soil types revealed distinct performance patterns. The model showed stable accuracy on chalky soils and high precision on clayey soils, while peat soils exhibited improved recall but reduced precision at higher BSI thresholds. Sandy soils were most sensitive to threshold variation, suggesting the potential need for localised calibration or feature refinement to maintain consistency across diverse geopedological settings.

B. Open Slurry Tank Detection Model

The Open Slurry Tank Detection Model was evaluated through two complementary validation strategies. First, augmentation-based validation applied geometric and photometric transformations, such as flips, rotations, brightness shifts, blur, noise, and compression artefacts, to test the model's robustness under varied visual conditions. Second, geographic generalisation validation was used to assess performance across unseen regions representing different farm typologies and landscape contexts. Across four augmentation iterations, the model achieved an average precision of 0.937, recall of 0.808, and mAP@0.5 of 0.880, demonstrating high consistency and resilience to image distortions. When tested on 233 diverse real-world locations, the model correctly detected nearly all true slurry tanks, with a recall of 98.26%, a precision of 71%, and an F1-score of 82.48%, confirming strong detection reliability and minimal missed positives. These findings indicate that the model performs effectively across varying geographies while sustaining high accuracy under practical imaging variations. The marginally lower precision of 71% indicates that the model has a slight tendency to misclassify similar structures as slurry tanks (i.e. generate false positives). However, this behaviour is preferred to higher rates of false negatives, as misclassified images can be manually corrected by user intervention.

C. Open Poached Land Detection Model

Validation was performed across 10 study tiles using visually confirmed poached sites derived from Google Earth Pro imagery. Evaluation metrics were computed at image level using manually labelled samples. Table 3 shows the overall accuracy matrices calculated for the model.

TABLE III: Overall accuracy matrices calculated for RDMAI Open Poached Land Detection Model

Metric (Average)	Value (%)
Overall Accuracy	65.7
Precision	62.2
Recall	59.3
F1-Score	60.7

Individual tile accuracy ranged between 44%–87%, reflecting spectral variability across soil types and moisture conditions. The model successfully distinguished compacted, disturbed soils with limited vegetation cover from bare cropland, though some confusion persisted in heterogeneous fields and semi-urban areas. Overall, model performance is moderate. However, considering that the model uses 10-m resolution satellite imagery to effectively identify small, granular disturbances in ground cover, it is an exciting area for future development.

V. CONCLUSION

Collectively, the Open Bare Cropland Detection Model, Open Slurry Tank Detection Model, and Open Poached Land Detection Model represent a significant advancement in geospatial intelligence for environmental monitoring. Used in

TABLE I: Geospatial data sources used to build the models

Datasets	Sources	Purposes
Sentinel 2 imagery (10 m)	Google Earth Engine	Computing spectral indices to create seasonal composites
Upscaled imagery (1 m)	S2DR3	Performing fine-scale image classification to detect small patches of degraded soil
Very High-Resolution Imagery (14 cm)	Commercial imagery	Validating model output as reference
Land use - Dynamic world	Google Earth Engine	Used in the Open Poached Land Detection Model to mask out built-up areas
Farmland	OpenStreetMap	Used in the Open Bare Cropland Detection Model to mask out non-crop fields
Farmyards	OpenStreetMap	Used to identify slurry tank locations for training of the Open Slurry Tank Detection Model
Roads	OpenStreetMap	Used in the Open Poached Land Detection Model to mask out roads from classified output
Crop type	CROME	Integrated with output of the Open Bare Cropland Detection Model to assess how specific crop types influence the risk of bare soil exposure
Rainfall	Defra Hydrology Explorer	Used in the Open Bare Cropland Detection Model to address issues with rainfall and bare soil detection
Catchment boundary	Defra/EA	Used in the application of the Open Bare Cropland Detection Model to targeted catchments

TABLE II: Accuracy Matrices for the RDMAI Open Bare Cropland Detection Model

Soil type	BSI Threshold > 0				BSI Threshold > 0.1			
	Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score
Chalky	0.94	0.93	0.97	0.95	0.94	0.99	0.91	0.95
Clayey	0.94	0.96	0.92	0.94	0.91	0.99	0.82	0.90
Peat	0.93	0.97	0.88	0.93	0.91	0.86	0.99	0.92
Sandy	0.94	0.92	0.92	0.92	0.87	0.98	0.71	0.83

conjunction, these three open-source tools form a comprehensive AI-driven framework capable of identifying both diffuse and point sources of agricultural pollution with unprecedented precision and transparency. Each model contributes a distinct analytical dimension:

- **Open Bare Cropland Detection Model:** maps exposed soils that heighten erosion and sediment runoff risk.
- **Open Slurry Tank Detection Model:** pinpoints agricultural circular slurry tanks that act as an indicator of both point and diffuse sources of agricultural pollution.
- **Open Poached Land Detection Model:** highlights compacted or disturbed soils that can accelerate overland flow and pollutant transport.

By combining these complementary perspectives, the framework enables regulators, researchers, and water companies to construct a multi-layered picture of catchment vulnerability to agricultural sources of water pollution, linking land-use dynamics with water quality outcomes. The use of cloud-based processing platforms (e.g., Google Earth Engine, Vertex AI) and open datasets (e.g., Sentinel-2, OpenStreetMap, Dynamic World) ensures scalability, repeatability, and accessibility across jurisdictions and institutions.

Beyond technical innovation, the initiative demonstrates the power of open collaboration in environmental data science. Each model is transparent, adaptable, and ready for integration into operational monitoring systems or future research pipelines. Together, they provide a foundation for proactive, data-driven management of agricultural landscapes, supporting the broader goals of sustainable land use, resilient ecosystems,

and cleaner rivers. As the models evolve through continued validation and integration with ground-based monitoring, they pave the way for a new era of evidence-based environmental governance, where machine learning, remote sensing, and open science converge to deliver actionable insights for a healthier planet.

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The model described in this document is the product of an ongoing research and development project. It represents an early iteration and its outputs are preliminary. The model is not intended for use in production environments, regulatory reporting, environmental monitoring, or any decision-making in relation to public health, public policy, or commercial matters without independent validation and the acceptance of full liability by the user.

All use of the model and its associated software is subject to the full terms set out in the RDMAI repository, available at <https://github.com/Cognizant-RDMAI>.

Readers seeking technical detail on model performance, limitations, and data sources should refer to the accompanying Model Report. The RDMAI consortium and its members accept no liability for any outcomes arising from use of or reliance on the model or its outputs.